

AURELIA PALACE

COMPLETE TECHNICAL REFERENCE & ARCHITECTURE GUIDE

1. Project Purpose & Scope

Aurelia Palace is an enterprise-standard full-stack web application designed for a luxury event venue. The system serves two distinct user personas: prospective clients exploring spaces, services, and submitting event enquiries; and internal event managers managing lead pipelines, updating enquiry statuses, and inspecting conversion statistics.

2. System Architecture & Cloud Infrastructure

The project implements a decoupled monorepo architecture where the Single Page Application (SPA) frontend and the REST API backend operate as independent services communicating over authenticated HTTPS JSON protocols:

- **Frontend Layer (Vercel):** React 18 + Vite SPA deployed on Vercel's global Edge CDN network for low-latency asset delivery.
- **Backend Layer (Render):** Node.js + Express 5 REST API web service with continuous deployment from GitHub.
- **Database Layer (Aiven Cloud):** Managed MySQL 8.4 database with TLS encrypted connections and ACID transaction safety.
- **Email Notification Service (Resend):** HTTP REST transactional email engine for automated customer confirmation emails.

3. Frontend Engineering & Advanced Techniques

- **React 18 & Vite:** Component-driven architecture using functional components, custom hooks (e.g. useFetch), and optimized production tree-shaking.
- **Dynamic SEO Management (react-helmet-async):** Every route dynamically updates page titles, meta descriptions, canonical links, and Open Graph social sharing cards.
- **Editorial Typography & Design Tokens:** Pairing of Cormorant Garamond serif and Inter sans-serif with custom Tailwind CSS color palettes (Charcoal, Stone, Ivory, Gold).
- **Micro-Interactions (Framer Motion):** Viewport-triggered fade-up reveals, smooth navigation menu transitions, interactive hover zoom states, and page unmounting animations.
- **Dual-Layer Validation (React Hook Form + Zod):** Client-side schema validation providing immediate user feedback on required fields, email formats, and guest minimums.
- **Axios Interceptors:** Request interceptors automatically inject JWT tokens for admin routes; response interceptors catch 401 Unauthorized errors and cleanly redirect to login.
- **Loading & Error Handling:** Custom Skeleton shimmer loaders prevent layout shifts during fetches, accompanied by dedicated full-page and inline error boundaries.
- **Masonry Gallery & Lightbox:** Multi-column responsive gallery with category filtering and interactive full-screen image preview lightbox.

4. Backend Architecture & Security Practices

- **Layered Controller-Service Architecture:** Strict separation between Express routing, Zod request body validation, controllers, and Prisma business logic services.
- **Defense-in-Depth Rate Limiting:** Tiered protection using express-rate-limit: Global API ceiling (100 req / 15 min), Auth protection (10 attempts / hr), and Enquiry protection (5 submissions / hr per IP).
- **Security Hardening (Helmet & CORS):** HTTP security headers for XSS, clickjacking, and content sniffing protection, with strict origin whitelisting matching the Vercel domain.
- **Stateless JWT Authentication:** Cryptographically signed JSON Web Tokens for administrative authorization without server-side session overhead.
- **Bcrypt Password Hashing:** Salted and hashed passwords ensuring admin credentials are never stored in plaintext.

- **Leak-Proof Centralized Error Handler:** Sanitized 500 responses in production to prevent leaking internal database schemas, credentials, or stack traces.
- **Structured Production Logging:** Morgan logging configured in Apache Combined format for cloud monitoring and anomaly detection.

5. Database Design & Relational Models (MySQL / Prisma)

The relational schema is managed using Prisma ORM 5.22 on MySQL 8.4, comprising 7 dedicated tables:

Model	Type / Purpose	Key Schema Attributes & Relationships
EventType	Venue Event Categories	name, slug (unique), description, image, featured (boolean), 1-to-many relation with Enquiry & GalleryItem.
Space	Venue Locations & Halls	name, slug (unique), capacity, description, image, features (JSON array), galleryImages (JSON array).
Service	Hospitality Offerings	name, slug (unique), category (indexed: Catering, Decor, Tech, Concierge), description, image, active flag.
GalleryItem	Curated Photography	title, category (indexed), image URL, caption, optional foreign key relation to EventType.
Story	Editorial Journal & Spotlights	title, slug (unique), excerpt, content (LongText markdown), coverImage, publishedDate, published flag.
Enquiry	Customer Lead Capture	name, email, phone, eventDate, guestCount, message, foreign key eventId, status (enum: NEW, CONTACTED, CLOSED).
AdminUser	Staff Authentication	name, email (unique), passwordHash (bcrypt encrypted), timestamps for audit tracking.

6. Key Business & Technical Workflows

A. Customer Enquiry & Fault-Tolerant Email Flow

- Client inputs details into the Enquiry form with live Zod client validation.
- Form dispatches POST /api/enquiries to the backend API.
- Server validates types and bounds before executing `prisma.enquiry.create()` into MySQL (Status: NEW).
- Database write completes as the immutable source of truth.
- Server queries the corresponding EventType name and triggers the Resend API to dispatch a branded HTML confirmation email.
- Fault Tolerance Guarantee:** If the email API encounters rate limits or network issues, the error is safely caught and logged server-side without rolling back or deleting the saved enquiry.
- Response returns 201 Created with an emailSent status flag, driving a dynamic frontend success notification.

B. Admin Authentication & Lead Lifecycle Flow

- Admin submits credentials on /admin/login; server verifies password hash via `bcrypt.compare()`.
- Server issues a signed JWT token; frontend persists it in localStorage.
- Admin dashboard loads summary statistics (/api/admin/stats) and paginated leads (/api/admin/enquiries).
- Admin can search leads by customer name/email and filter by status.
- Lead status is updated via PATCH /api/admin/enquiries/:id/status through an enum transition (NEW → CONTACTED → CLOSED).

7. Cloud DevOps & Deployment Architecture

- Frontend Edge Hosting (Vercel):** Direct GitHub repository integration triggers automated builds (vite build) on pushes to main, distributing assets globally across edge servers.
- Backend Web Service (Render):** Managed Node.js service executing continuous deployment with automatic zero-downtime health verification.
- Database Cloud Service (Aiven):** High-availability MySQL 8.4 instance configured with automated backups and secure TLS connections.
- Environment Secrets Isolation:** Strict isolation of DATABASE_URL, JWT_SECRET, RESEND_API_KEY, and CLIENT_URL through environment variables excluded from Git version control.